

Software Test Plan (STP)

Pocket Packet Sniffer

Project	Pocket Packet Sniffer
Version	1.3
Authors	Vikas Kulakarni, Shivanand, Shivakumara GP, Sanjith R
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1. Introduction

Purpose: This document defines the test plan for Pocket Packet Sniffer v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities.

Scope: Testing covers packet capture, packet parsing and classification, capture storage and query, live monitoring, report export, and stretch features such as suspicious activity detection and multi-backend capture.

References: Pocket Packet Sniffer SRS v1.3, Project Proposal, Scapy documentation, libpcap documentation.

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), pcap (Packet Capture), CLI (Command Line Interface), RTM (Requirements Traceability Matrix).

2. Test Items

- Packet Capture module
- Packet Parsing & Classification module
- Capture Storage & Query module
- Live Monitoring Dashboard
- Suspicious Activity Detection (Stretch)
- Multi-Backend Capture (Stretch)

3. Features to be Tested

Requirement	Feature
REQ-1	Capture raw packets
REQ-2	Interface/permission error handling
REQ-3	Extract packet metadata
REQ-4	Add protocol analyzers
REQ-5	Store captures
REQ-6	Filter/query captures
REQ-7	Display live statistics
REQ-8	Export session report
REQ-9	Detect suspicious activity
REQ-10	Multiple backend capture

4. Features Not to be Tested

- Unauthorized network traffic
- Packet injection, modification, or replay
- Internal implementation of third-party libraries

- BLE and multi-backend features if not implemented as stretch goals
- Web dashboard if not implemented

5. Test Approach / Strategy

Levels: Unit testing, Integration testing, System testing, Acceptance testing.

Types: Functional testing, Regression testing, Performance testing, Usability testing, Security testing.

Entry Criteria: Stable build, test environment, required interface, and test data available.

Exit Criteria: All planned test cases executed, critical defects resolved, and requirements verified.

5.1 Security Validation

- Verify authorized network capture
- Verify capture file permissions
- Verify no silent data upload
- Verify no packet injection/modification/replay

6. Test Environment

Hardware: Raspberry Pi, Ethernet NIC, WiFi adapter supporting monitor mode.

Software: Linux/Raspberry Pi OS, Python 3.x, Scapy, libpcap, SQLite.

Tools: PyTest, Scapy, Wireshark, Git.

Test Data: TCP, UDP, ARP, ICMP and other authorized network traffic.

7. Test Schedule

Activity	Planned Period
Test case design	Week 1–2
Environment setup	Week 2
Unit testing	Week 3–4
Integration testing	Week 5
System testing	Week 6
Final testing & documentation	Week 7

8. Test Deliverables

- Test Plan
- Test Cases
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- RTM
- Test Summary Report

9. Roles and Responsibilities

Role	Responsibility
Vikas Kulakarni	Capture testing

Shivanand	Parsing & classification testing
Shivakumara GP	Storage & query testing
Sanjith R	Dashboard & presentation testing
All Members	Integration testing and final review

10. Risks and Mitigation

Risk	Mitigation
WiFi monitor mode unavailable	Use Ethernet backend
BLE implementation delayed	Keep as stretch goal
Layer boundaries become unclear	Review interfaces regularly
Sensitive captured traffic	Use authorized networks and secure files

11. Assumptions & Dependencies

- Raspberry Pi or similar hardware is available.
- Required network interfaces are available.
- Scapy/libpcap are compatible with the environment.
- Test traffic is generated only on authorized networks.
- Required permissions are available.

12. Suspension & Resumption Criteria

Suspend testing if: Test environment is unavailable, build is unstable, or critical defects block testing.

Resume testing if: Blocking defects are resolved, environment is restored, and a stable build is available.

13. Test Case Management & Traceability

Requirement	Test Case
REQ-1	TC-CAP-01
REQ-2	TC-CAP-02
REQ-3	TC-ANA-01
REQ-4	TC-ANA-02
REQ-5	TC-STO-01
REQ-6	TC-STO-02
REQ-7	TC-PRE-01
REQ-8	TC-PRE-02
REQ-9	TC-SEC-01
REQ-10	TC-CAP-03

14. Test Metrics & Reporting

Metrics: Test cases executed, pass/fail percentage, requirement coverage, defect count, defect resolution, packet loss, dashboard refresh time.

Reports: Test execution report, defect report, final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
Team Lead		
QA/Test Lead		
Faculty/Instructor		